HAMZA QURESHI
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SUMMARY
AI graduate who ships LLM features end to end: data pipelines, retrieval, APIs and deployment.

EDUCATION
BS Artificial Intelligence, Indus Valley University of Technology (fictional), 2022 - 2026. CGPA 3.7/4.0
Coursework: Machine Learning, Deep Learning, NLP, Database Systems, Software Engineering.

EXPERIENCE
AI Engineering Intern, Kestrel Data Works (fictional), Jun 2025 - Sep 2025
- Built a RAG assistant over 2,000 internal support tickets using LangChain and pgvector in PostgreSQL.
- Wrote SQL queries with joins and window functions to build the evaluation dataset of 400 question-answer pairs.
- Exposed the assistant through a FastAPI REST service with request validation using Pydantic.
- Packaged the service in Docker and deployed it on a Google Cloud Run instance.

Teaching Assistant, Machine Learning course, Indus Valley University of Technology, Feb 2025 - Jun 2025
- Ran weekly labs for 60 students on scikit-learn pipelines, cross-validation and overfitting.

PROJECTS
Urdu News Classifier (Final Year Project)
- Fine-tuned a multilingual BERT model with Hugging Face Transformers on 18,000 Urdu news articles; 89% macro F1.
- Tracked experiments in MLflow and compared against a TF-IDF + logistic regression baseline (81% F1).

Resume Q&A Bot
- Built a prompt-engineered chatbot with the Groq API and a Chroma vector store; added citation checks to reduce hallucinations.

Traffic Sign Detector
- Trained a CNN in PyTorch with data augmentation on 30,000 images (94% test accuracy).

SKILLS
Python, SQL, PyTorch, scikit-learn, pandas, NumPy, Hugging Face Transformers, LangChain, FastAPI, Docker, Git, GitHub Actions, PostgreSQL

CERTIFICATIONS
DeepLearning.AI Machine Learning Specialization (completed)

ADDITIONAL PROJECTS
Project 1: Inventory Insights v1 (fictional client)
- Built a Python ETL job that loads 1000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 4 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 2: Inventory Insights v2 (fictional client)
- Built a Python ETL job that loads 2000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 5 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 3: Inventory Insights v3 (fictional client)
- Built a Python ETL job that loads 3000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 6 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 4: Inventory Insights v4 (fictional client)
- Built a Python ETL job that loads 4000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 7 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 5: Inventory Insights v5 (fictional client)
- Built a Python ETL job that loads 5000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 8 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 6: Inventory Insights v6 (fictional client)
- Built a Python ETL job that loads 6000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 9 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 7: Inventory Insights v7 (fictional client)
- Built a Python ETL job that loads 7000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 10 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 8: Inventory Insights v8 (fictional client)
- Built a Python ETL job that loads 8000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 11 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 9: Inventory Insights v9 (fictional client)
- Built a Python ETL job that loads 9000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 12 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 10: Inventory Insights v10 (fictional client)
- Built a Python ETL job that loads 10000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 13 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 11: Inventory Insights v11 (fictional client)
- Built a Python ETL job that loads 11000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 14 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 12: Inventory Insights v12 (fictional client)
- Built a Python ETL job that loads 12000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 15 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 13: Inventory Insights v13 (fictional client)
- Built a Python ETL job that loads 13000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 16 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 14: Inventory Insights v14 (fictional client)
- Built a Python ETL job that loads 14000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 17 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 15: Inventory Insights v15 (fictional client)
- Built a Python ETL job that loads 15000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 18 recurring bugs.
- Presented the results to the client in a 15-minute demo.

Project 16: Inventory Insights v16 (fictional client)
- Built a Python ETL job that loads 16000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 19 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 17: Inventory Insights v17 (fictional client)
- Built a Python ETL job that loads 17000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 20 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 18: Inventory Insights v18 (fictional client)
- Built a Python ETL job that loads 18000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 21 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 19: Inventory Insights v19 (fictional client)
- Built a Python ETL job that loads 19000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 22 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 20: Inventory Insights v20 (fictional client)
- Built a Python ETL job that loads 20000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 23 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 21: Inventory Insights v21 (fictional client)
- Built a Python ETL job that loads 21000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 24 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 22: Inventory Insights v22 (fictional client)
- Built a Python ETL job that loads 22000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 25 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 23: Inventory Insights v23 (fictional client)
- Built a Python ETL job that loads 23000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 26 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 24: Inventory Insights v24 (fictional client)
- Built a Python ETL job that loads 24000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 27 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 25: Inventory Insights v25 (fictional client)
- Built a Python ETL job that loads 25000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 28 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 26: Inventory Insights v26 (fictional client)
- Built a Python ETL job that loads 26000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 29 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 27: Inventory Insights v27 (fictional client)
- Built a Python ETL job that loads 27000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 30 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 28: Inventory Insights v28 (fictional client)
- Built a Python ETL job that loads 28000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 31 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 29: Inventory Insights v29 (fictional client)
- Built a Python ETL job that loads 29000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 32 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 30: Inventory Insights v30 (fictional client)
- Built a Python ETL job that loads 30000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 33 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 31: Inventory Insights v31 (fictional client)
- Built a Python ETL job that loads 31000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 34 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 32: Inventory Insights v32 (fictional client)
- Built a Python ETL job that loads 32000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 35 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 33: Inventory Insights v33 (fictional client)
- Built a Python ETL job that loads 33000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 36 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 34: Inventory Insights v34 (fictional client)
- Built a Python ETL job that loads 34000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 37 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 35: Inventory Insights v35 (fictional client)
- Built a Python ETL job that loads 35000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 38 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 36: Inventory Insights v36 (fictional client)
- Built a Python ETL job that loads 36000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 39 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 37: Inventory Insights v37 (fictional client)
- Built a Python ETL job that loads 37000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 40 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 38: Inventory Insights v38 (fictional client)
- Built a Python ETL job that loads 38000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 41 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 39: Inventory Insights v39 (fictional client)
- Built a Python ETL job that loads 39000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 42 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 40: Inventory Insights v40 (fictional client)
- Built a Python ETL job that loads 40000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 43 recurring bugs.

- Presented the results to the client in a 15-minute demo.
Project 41: Inventory Insights v41 (fictional client)
- Built a Python ETL job that loads 41000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 44 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 42: Inventory Insights v42 (fictional client)
- Built a Python ETL job that loads 42000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 45 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 43: Inventory Insights v43 (fictional client)
- Built a Python ETL job that loads 43000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 46 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 44: Inventory Insights v44 (fictional client)
- Built a Python ETL job that loads 44000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 47 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 45: Inventory Insights v45 (fictional client)
- Built a Python ETL job that loads 45000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 48 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 46: Inventory Insights v46 (fictional client)
- Built a Python ETL job that loads 46000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 49 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 47: Inventory Insights v47 (fictional client)
- Built a Python ETL job that loads 47000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 50 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 48: Inventory Insights v48 (fictional client)
- Built a Python ETL job that loads 48000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 51 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 49: Inventory Insights v49 (fictional client)
- Built a Python ETL job that loads 49000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 52 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 50: Inventory Insights v50 (fictional client)
- Built a Python ETL job that loads 50000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 53 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 51: Inventory Insights v51 (fictional client)
- Built a Python ETL job that loads 51000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 54 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 52: Inventory Insights v52 (fictional client)
- Built a Python ETL job that loads 52000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 55 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 53: Inventory Insights v53 (fictional client)
- Built a Python ETL job that loads 53000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 56 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 54: Inventory Insights v54 (fictional client)
- Built a Python ETL job that loads 54000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 57 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 55: Inventory Insights v55 (fictional client)
- Built a Python ETL job that loads 55000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 58 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 56: Inventory Insights v56 (fictional client)
- Built a Python ETL job that loads 56000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 59 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 57: Inventory Insights v57 (fictional client)
- Built a Python ETL job that loads 57000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 60 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 58: Inventory Insights v58 (fictional client)
- Built a Python ETL job that loads 58000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 61 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 59: Inventory Insights v59 (fictional client)
- Built a Python ETL job that loads 59000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 62 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 60: Inventory Insights v60 (fictional client)
- Built a Python ETL job that loads 60000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 63 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 61: Inventory Insights v61 (fictional client)
- Built a Python ETL job that loads 61000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 64 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 62: Inventory Insights v62 (fictional client)
- Built a Python ETL job that loads 62000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 65 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 63: Inventory Insights v63 (fictional client)
- Built a Python ETL job that loads 63000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 66 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 64: Inventory Insights v64 (fictional client)
- Built a Python ETL job that loads 64000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 67 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 65: Inventory Insights v65 (fictional client)
- Built a Python ETL job that loads 65000 rows per day into PostgreSQL.

- Wrote a short report on data quality issues and fixed 68 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 66: Inventory Insights v66 (fictional client)
- Built a Python ETL job that loads 66000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 69 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 67: Inventory Insights v67 (fictional client)
- Built a Python ETL job that loads 67000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 70 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 68: Inventory Insights v68 (fictional client)
- Built a Python ETL job that loads 68000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 71 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 69: Inventory Insights v69 (fictional client)
- Built a Python ETL job that loads 69000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 72 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 70: Inventory Insights v70 (fictional client)
- Built a Python ETL job that loads 70000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 73 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Project 71: Inventory Insights v71 (fictional client)
- Built a Python ETL job that loads 71000 rows per day into PostgreSQL.
- Wrote a short report on data quality issues and fixed 74 recurring bugs.
- Presented the results to the client in a 15-minute demo.
Configured Kubernetes autoscaling for the Falcon Freight (fictional) tracking API.
Volunteer work 72: taught a weekend coding class for school students (cohort 72).
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Volunteer work 74: taught a weekend coding class for school students (cohort 74).
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Volunteer work 76: taught a weekend coding class for school students (cohort 76).
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Volunteer work 80: taught a weekend coding class for school students (cohort 80).
Volunteer work 81: taught a weekend coding class for school students (cohort 81).
Volunteer work 82: taught a weekend coding class for school students (cohort 82).
Volunteer work 83: taught a weekend coding class for school students (cohort 83).
Volunteer work 84: taught a weekend coding class for school students (cohort 84).
Volunteer work 85: taught a weekend coding class for school students (cohort 85).
Volunteer work 86: taught a weekend coding class for school students (cohort 86).
Volunteer work 87: taught a weekend coding class for school students (cohort 87).
Volunteer work 88: taught a weekend coding class for school students (cohort 88).
Volunteer work 89: taught a weekend coding class for school students (cohort 89).
Volunteer work 90: taught a weekend coding class for school students (cohort 90).
Volunteer work 91: taught a weekend coding class for school students (cohort 91